

ENVS-193DS Final

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Table of contents

Problem 1. Research writing	2
Problem 2. Data Analysis	2
a.	2
b.	3
c.	4
Tree Height (height_cm)	4
Acacia ant species (ant_species)	4
d.	4
e.	4
f.	4
g.	5
h.	5
i.	7
j.	8
k.	8
Problem 3. Problem 3. Affective and exploratory visualizations	8
a.	8
b.	8
c.	9
Checking assumptions	9
Running analysis	10
d.	10
e.	12
f.	12

[Link to repo](#)

```

# read in packages
library(tidyverse)
library(here)
library(DHARMA)
library(ggeffects)
library(rstatix)
library(janitor)

# reading in data and creating new object
nest <- read_csv(
  here("data", "nest_data_final.csv"))

# creating new data frame with personal data
personal_data <- read_csv(here("data", "personal-datasheet.csv")) |>
  # standardizing column names
  clean_names() |>
  # dropping rows with na observations
  drop_na() |>
  # changing the stop column to display TRUE/FALSE instead of 1/0
  mutate(stop = as.logical(stop),
         week = as.factor(week))

```

Problem 1. Research writing

Problem 2. Data Analysis

a.

```

# creating clean data frame
nests_clean <- nest |>
  # selecting desired columns
  select(ant_species, height_cm, case_control) |>
  # filtering for desired ant species
  filter(ant_species %in% c("RRB", "BBR", "AB")) |>
  # creating new column for full scientific name
  mutate(sci_name = case_when(
    ant_species == "RRB" ~ "Crematogaster sjostedti",
    ant_species == "BBR" ~ "Crematogaster mimosae",
    ant_species == "AB" ~ "Crematogaster nigriceps"),

```

```

# reordering factors for ant species
ant_species = factor(ant_species,
  levels = c("RRB", "BBR", "AB"))
)

```

```

# displaying structure of data frame
str(nests_clean)

```

```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ ant_species : Factor w/ 3 levels "RRB","BBR","AB": 1 1 1 1 3 1 2 1 3 2 ...
 $ height_cm   : num [1:270] 392 310 513 154 324 ...
 $ case_control: num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ sci_name    : chr [1:270] "Crematogaster sjostedti" "Crematogaster sjostedti" "Crematogaster sjostedti" ...

```

```

# slicing sample
slice_sample(nests_clean, n = 10)

```

```

# A tibble: 10 x 4
  ant_species height_cm case_control sci_name
  <fct>         <dbl>         <dbl> <chr>
1 RRB           129             0 Crematogaster sjostedti
2 BBR           318.            1 Crematogaster mimosae
3 BBR           155             0 Crematogaster mimosae
4 AB            130.            0 Crematogaster nigriceps
5 BBR           170             0 Crematogaster mimosae
6 RRB           264.            0 Crematogaster sjostedti
7 RRB           191             1 Crematogaster sjostedti
8 RRB           147             0 Crematogaster sjostedti
9 RRB           187             0 Crematogaster sjostedti
10 BBR          166.            1 Crematogaster mimosae

```

b.

For this data, a value of 1 in the case_control column indicates that a nest was observed in the acacia tree in question. A value of zero in that said column means that no bird nest was observed in the tree.

c.

Tree Height (height_cm)

Tree height in this data set is a continuous variable that is measured in centimeters. As tree height increases the probability of nest occurrence would likely increase. Taller trees are more structurally complex and therefore have more nesting opportunities, they may also provide better predator avoidance for birds (Introduction, paragraph 2; Methods, “Study site and species” subsection, paragraph 1).

Acacia ant species (ant_species)

As ant species changes, the probability of nest occurrence would likely differ. This is due to each ant species exhibiting a distinct aggression level and mutualistic relationship with the host Acacia tree which affects the level of predator protection a nest in said tree would receive. (Introduction, paragraph 3; Methods, “Nest box experiment” subsection, paragraph 1)

d.

Model number	Ant Species	Tree Height	Predictor list
1	X	X	all predictors, no interaction (null model)
2	X	X	all predictors, with interaction (full model)

e.

f.

```
# running AIC test to determine best model
AIC(model1, model2)
```

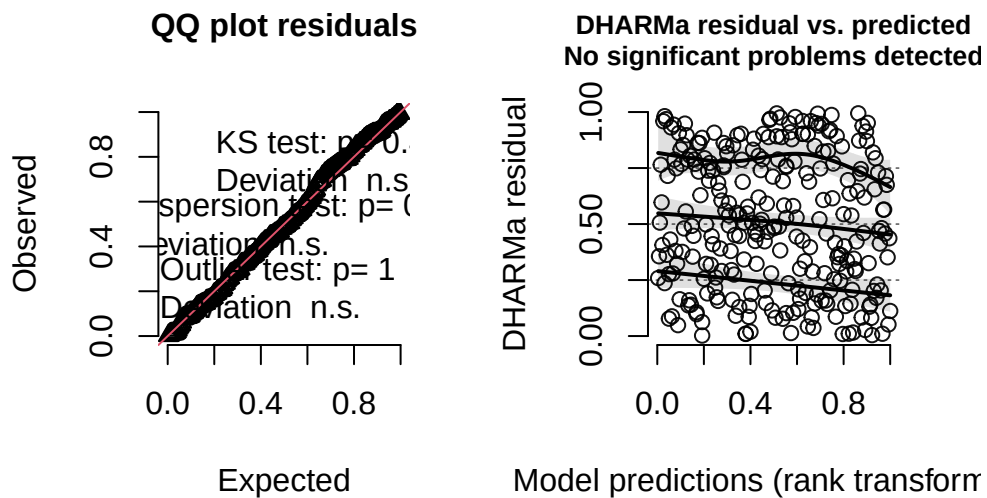
```
      df      AIC
model1  4 258.7430
model2  6 237.8042
```

The best model as determined by Akaike’s Information Criterion (AIC) was the model that predicted test_case (whether or not a sample tree had a nest) using the interaction of tree height (height_cm) and species of acacia ant (ant_species)

g.

```
# plotting simulated residuals from DHARMA package.  
plot(  
  simulateResiduals(model2)  
)
```

DHARMA residual



h.

```
# creating prediction model for model2  
preds <- ggpredict(model2,  
  # terms for model, all tree heights all species  
  terms = c("height_cm[all]", "ant_species")) |>  
# changing grouping column's name  
rename(ant_species = group) |>  
arrange(x)  
  
# setting custom species colors  
sp_cols <- c(  
  "RRB" = "tan",
```

```

"BBR" = "orange",
"AB" = "lightblue3"
)

# creating facet labels from nest_clean
sp_labels <- nests_clean |>
  # choosing correct columns
  distinct(ant_species, sci_name) |>
  deframe()

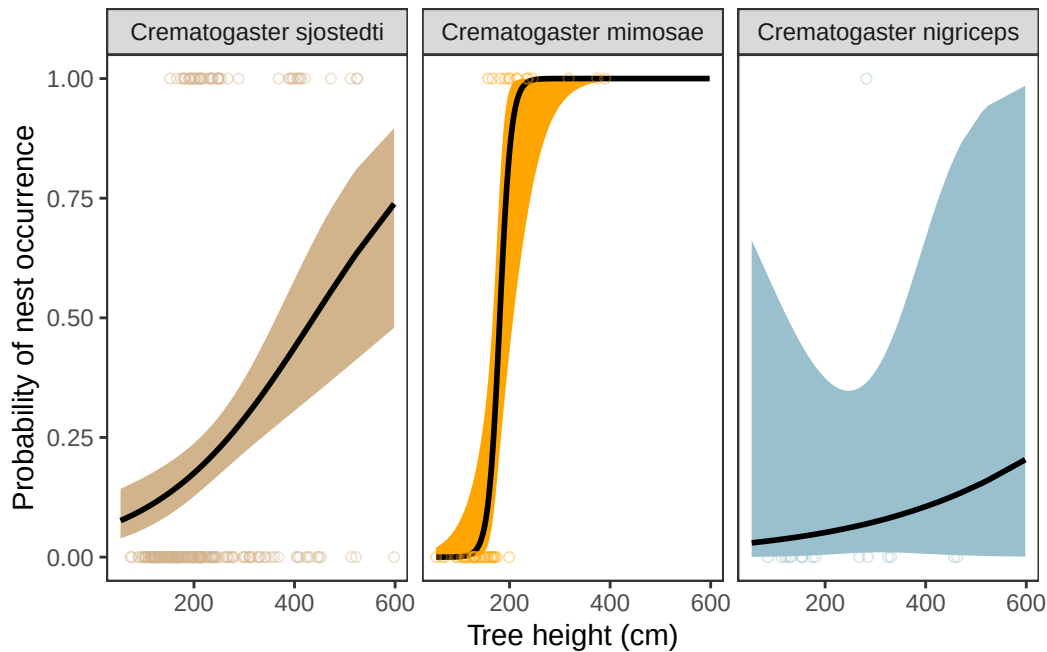
# base layer: ggplot
ggplot() +
  # first layer: model confidence ribbon
  geom_ribbon(data = preds, # prediction data frame
             aes(x = x,
                 ymin = conf.low, # min by low confidence
                 ymax = conf.high, # max by high confidence
                 fill = ant_species)) + # filling color by group
  # second layer: model probability line
  geom_line(data = preds, # prediction data frame
            aes(x = x,
                y = predicted),
            # coloring line black
            color = "black",
            # making line thicker
            linewidth = 1) +
  # third layer: jitter for underlying data
  geom_jitter(data = nests_clean, # nests_clean data frame
              aes(x = height_cm, # x-axis: tree height
                  y = case_control, # y-axis: nest occurrence
                  color = ant_species), # coloring by species
              alpha = 0.35, # making points more opaque
              shape = 21, # changing shape
              height = 0) + # removing vertical jitter
  # separating by species
  facet_wrap(~ ant_species,
             # labelling with sci_name
             labeller = labeller(ant_species = sp_labels)) +
  # setting colors by species
  scale_color_manual(values = sp_cols) +
  scale_fill_manual(values = sp_cols) +
  # relabelling x and y-axes

```

```

labs(
  x = "Tree height (cm)",
  y = "Probability of nest occurrence"
) +
# changing theme
theme_bw()+
# removing legend
theme(legend.position = "none",
      # removing gridlines
      panel.grid = element_blank())

```



i.

Figure 1. Model predictions for bird nest occurrence in acacia trees based on height and ant species Figure displays model predictions for probability of nest occurrence as tree height increases, separated by ant species living on the tree. Line represents exact predictions of model that assumes interaction between species and tree height. Ribbon represents model 95% confidence interval. Species are sorted by color with *C. sjostedti* being tan, *C. mimosae* being orange, and *C. nigriceps* being blue. Data originally from Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j.

```
# creating prediction at 600 cm
predict_600 <- ggpredict(model2,
                          terms = c("height_cm [600]", "ant_species"))
```

k.

Problem 3. Problem 3. Affective and exploratory visualizations

a.

- My affective visualization is pretty much a completely new way to display data when compared to my initial exploratory plots. In effect, it is still a sort of scatter plot but point shape and color matter far more than location. There is also no models or metrics of central tendency as my goal with the affective visualization was not to make something that explained the whole dataset.
- Honestly it is pretty much impossible to see any similarities between my plots and my affective piece, they share no discernible characteristics besides the underlying data. I did separate by location in each case but I did so differently each time.
- Between the two exploratory plots there are some trends, mainly that walk time doesn't change too much on average between locations. There is also a pretty clear negative relationship between time left and total walk time that can be seen without any models.
- The main feedback I got was on improving readability. My visual was hard to understand but with time and explanation it became more legible for my peers. The visual improvements and readability improvements I made, upping line thickness, marking stops, using one line, etc., were all based on this conversation with my peers.

b.

Using walk_minutes as a response and week as the predictor I believe the most interesting analysis we can run is a t-test (or Mann-Whitney U if nonparametric) to see if mean (or median) walk time differs across location.

This is important to my central question of what is effecting my walk time to class, seeing if there is a difference across location could potentially invalidate my previous work on this data frame where I assumed non-differing central tendency between the two locations allowing for both locations' data to be used together.

Walk time is a continuous numeric variable.

Location is a categorical variable with 2 levels: Campbell (n = 19) and 387 (n = 13)

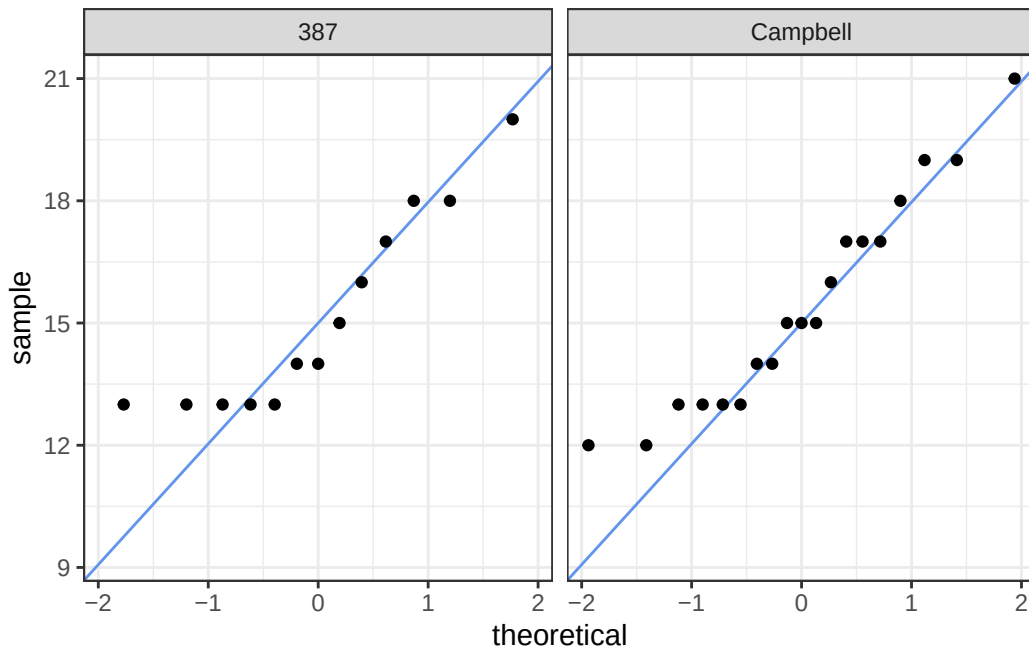
c.

Checking assumptions

- Observations are independent
- Response variable is continuous

Running QQ to test normality

```
# baselayer: ggplot
ggplot(data = personal_data, # using urchins clean data frame
       aes(sample = walk_minutes)) + # y-axis for QQ plot
# first layer: QQ reference line
geom_qq_line(color = "cornflowerblue") + # displaying line in blue
# second layer: QQ plot
geom_qq() +
# creating facets
facet_wrap(~location) +
# changing theme
theme_bw()
```



Not very normal looking, some grouping along line but likely need to use non-parametric due to low sample size.

Running analysis

```
wilcox_test(walk_minutes ~ location,  
            data = personal_data)
```

```
# A tibble: 1 x 7  
  .y.      group1 group2      n1      n2 statistic      p  
* <chr>      <chr> <chr>    <int> <int>    <dbl> <dbl>  
1 walk_minutes 387   Campbell    13    19     118.  0.83
```

Interpretation: Mann-Whitney test, $U = 118$, $p = 0.83$, $\alpha = 0.05$

No significant difference in rank distributions.

d.

```
location_medians <- personal_data |>  
  group_by(location) |>  
  summarise(median_walk = median(walk_minutes),  
            q25 = quantile(walk_minutes, 0.25),  
            q75 = quantile(walk_minutes, 0.75)) |>  
  ungroup()  
  
ggplot(data = personal_data,  
       aes(x = location,  
           y = walk_minutes)) +  
  geom_rect(data = location_medians,  
            aes(xmin = as.numeric(factor(location)) - 0.25,  
                xmax = as.numeric(factor(location)) + 0.25,  
                ymin = q25,  
                ymax = q75,  
                x = NULL,  
                y = NULL),  
            fill = "#003090",  
            alpha = 0.15) +  
  geom_crossbar(data = location_medians,
```

[illegible]

e.

Figure 2. No significant difference in walk time distributions across locations. Blue cross bars represent median walk time to class for each location (Campbell $n = 19$ and 387 $n = 13$) with the rectangle representing the IQR for each week's observations. Underlying data is shown via the orange points which each represent a single day's walk time. Data collected by Luke Simon between the dates of 2026-04-20 and 2026-06-03.

f.

From the dates of 2026-04-20 and 2026-06-03 there is no difference in daily walk time distributions between building 387 (median = 14, $n = 13$) and Campbell Hall (median = 15, $n = 19$) (Mann-Whitney test, $U = 118$, $p = 0.83$, $\alpha = 0.05$). Since our result is not significant we can assume that distributions are nearly identical (as is represented in the above figure) meaning that all previous analysis done and linear models created that do not differentiate between location are still valid. Essentially, walk times to each location are comparable and can be treated as independent observations in the same data set, any differentiation between the two locations would be included for clarity and aesthetic purposes only.